

Figure Provenance & Pipeline Consistency Audit

Date: 2026-03-21
Scope: docs/**/* .md , README.md , results/**
Mode: READ-ONLY — No files modified

1. FIGURE PROVENANCE TABLE

All `![...](/home/ubuntu/ems-optimization/...)` references extracted from non-archive markdown files (excluding badge shields and docs/archive/).

#	Figure Path (resolved)	Used In (Doc)	Tier	Script Source	Exists?	Status
1	results/ baseline/ figures/ policy_com paris- on_panel_K 20_cap2.png	docs/core/ technical_report. md:433 (Figure 1)	baseline	Unknown — no script found that generates this file-name	✓ Yes (642 KB)	⚠ No generating script identified
2	results/ baseline/ figures/ response_time_distribution_by_policy.png	docs/core/ technical_report. md:439 (Figure 2)	baseline	scripts/ analysis/ regenerate_all_figures.py → fig2_rt_distribution()	✓ Yes (340 KB)	⚠ Script writes to results/figures/ , not results/baseline/figures/
3	results/ archive/ figures/ fleet_sensitivity_dual.png	docs/core/ technical_report. md:478 (Figure 3)	archive	scripts/ analysis/ regenerate_all_figures.py → fig3_fleet_sensitivity()	✓ Yes (610 KB)	⚠ Archive figure; script writes to results/figures/ ; baseline has fleet_sensitivity_v2_dual.png (182 KB, different hash)
4	results/ analysis/ figures/ cbd_robustness_enhanced.png	docs/core/ technical_report. md:536 (Figure 4)	analysis	scripts/ analysis/ regenerate_all_figures.py → fig4_cbd_r	✓ Yes (452 KB)	⚠ Script reads from results/simulation/cbd_experiment/ which

#	Figure Path (resolved)	Used In (Doc)	Tier	Script Source	Exists?	Status
				obustness()		does NOT exist; data is at results/analysis/simulation/cbd_experiment/
5	results/analysis/figures/queue_comparison_by_policy.png	docs/core/technical_report.md:559	analysis	scripts/analysis/analyze_queue_metrics.py	✓ Yes (171 KB)	⚠ Script writes to results/figures/ , not results/analysis/figures/
6	results/analysis/figures/seasonal_patterns.png	docs/core/technical_report.md:582	analysis	scripts/analysis/analyze_seasonal_patterns.py	✓ Yes (278 KB)	⚠ Script writes to results/figures/ , not results/analysis/figures/
7	results/analysis/distancemparison/distancemparison_bar.png	docs/core/technical_report.md:598	analysis	scripts/analysis/run_distance_comparison_experiment.py	✓ Yes (92 KB)	✓ Script writes to correct directory
8	results/analysis/figures/cbd_equity_tradeoff_	docs/core/technical_report.md:621 (Figure 5)	analysis	scripts/analysis/regenerate_all_figures.py	✓ Yes (392 KB)	⚠ Script writes to results/figures/ , not res-

#	Figure Path (resolved)	Used In (Doc)	Tier	Script Source	Exists?	Status
	summary.png			→ fig5_cbd_e quity()		ults/analysis/figures/
9	results/ archive/ figures/ capacity_sensitivity_heatmap.png	docs/core/ technical_report. md:670 (Figure 6)	archive	scripts/ analysis/ regenerate_all_figures.py → fig6_capacity_heatmap() OR scripts/ analysis/ generate_capacity_sensitivity_heatmap.py	✓ Yes (135 KB)	● CRITICAL: Both scripts read from results/capacity_comparison/ which does NOT exist; data is at results/analysis/capacity_comparison/. Figure is NOT reproducible. Known “Data format issue” bug was fixed but image in archive may be stale.
10	results/ analysis/ capacity_comparison/performance_vs_capacity-	docs/analysis/capacity_sensitivity_analysis.md:325	analysis	scripts/ analysis/ capacity_sensitivity_full_spectrum.py	✓ Yes (283 KB)	✓ OK

#	Figure Path (resolved)	Used In (Doc)	Tier	Script Source	Exists?	Status
	city_K40.png					
11	results/analysis/capacity_comparison/tradeoff_distribution_rt_K40.png	docs/analysis/capacity_sensitivity_analysis.md:328	analysis	scripts/analysis/capacity_sensitivity_full_spectrum.py	✓ Yes (105 KB)	✓ OK
12	results/analysis/capacity_comparison/rt_heatmap_K40.png	docs/analysis/capacity_sensitivity_analysis.md:331	analysis	scripts/analysis/capacity_sensitivity_full_spectrum.py	✓ Yes (59 KB)	✓ OK
13	results/analysis/capacity_comparison/full_spectrum_summary.png	docs/analysis/capacity_sensitivity_analysis.md:334	analysis	scripts/analysis/capacity_sensitivity_full_spectrum.py	✓ Yes (270 KB)	✓ OK
14	results/analysis/figures/cbd_response_comparison.png	docs/analysis/cbd_robustness_analysis.md:83	analysis	notebooks/09_cbd_analysis.ipynb (cell output)	✓ Yes (183 KB)	⚠ Generated by notebook, not a standalone script
15	results/analysis/figures/cbd_scenar	docs/analysis/cbd_robustness_analysis.md:86	analysis	notebooks/09_cbd_analysis.ipynb	✓ Yes (233 KB)	⚠ Generated by notebook, not a stan-

#	Figure Path (resolved)	Used In (Doc)	Tier	Script Source	Exists?	Status
	io_comparison.png			(cell output)		dalone script
16	results/analysis/figures/cbd_heatmap.png	docs/analysis/cbd_robustness_analysis.md:89	analysis	notebooks/09_cbd_analysis.ipynb (cell output)	✓ Yes (235 KB)	⚠ Generated by notebook, not a standalone script
17	results/analysis/figures/queue_comparison_by_policy.png	docs/analysis/queue_analysis.md:105	analysis	scripts/analysis/analyze_queue_metrics.py	✓ Yes (171 KB)	⚠ Script output path mismatch (see #5)
18	results/analysis/figures/queue_vs_fleet_size.png	docs/analysis/queue_analysis.md:108	analysis	scripts/analysis/analyze_queue_metrics.py	✓ Yes (158 KB)	⚠ Same output path mismatch
19	results/analysis/figures/queue_vs_demand.png	docs/analysis/queue_analysis.md:111	analysis	scripts/analysis/analyze_queue_metrics.py	✓ Yes (170 KB)	⚠ Same output path mismatch
20	results/analysis/figures/queue_heatmap.png	docs/analysis/queue_analysis.md:114	analysis	scripts/analysis/analyze_queue_metrics.py	✓ Yes (361 KB)	⚠ Same output path mismatch

Summary: 19 unique figure references found. All 19 files exist. **0 missing files.** However, significant pipeline consistency issues exist.

2. DUPLICATE FIGURE GROUPS

Group A: Fleet Sensitivity

File	Location	Size	MD5
fleet_sensitivity_dual.png	results/archive/figures/	610 KB	959a8c02...
fleet_sensitivity_v2_dual.png	results/baseline/figures/	182 KB	cb5c9c5d...
fleet_sensitivity_archive.png	results/archive/figures/	67 KB	—
production_fleet_sensitivity.png	results/baseline/figures/	—	—
pub_fig2_fleet_sensitivity.png	results/baseline/figures/	—	—
exp2_fleet_sensitivity.png	results/baseline/figures/	—	—

Issue: The technical report (Figure 3) references the **archive** version (fleet_sensitivity_dual.png). A newer version exists in baseline/figures/ as fleet_sensitivity_v2_dual.png (different content, different hash). The archive version is 3.3× larger. **The report may be referencing an outdated figure.**

Group B: Capacity Sensitivity Heatmap

File	Location	Size
capacity_sensitivity_heatmap.png	results/archive/figures/	135 KB
capacity_sensitivity_heatmap_notebook.png	results/analysis/figures/	—

Issue: The technical report (Figure 6) references the **archive** version. The user-uploaded screenshot shows this figure with “K=20 Data format issue” and “K=40 Data format issue” text — confirming the archive figure is the **broken** version. A notebook-generated alternative exists but is not referenced.

Group C: Policy Comparison

File	Location
fig_policy_comparison.png	results/baseline/figures/
policy_comparison.png	results/baseline/figures/
pub_fig1_policy_comparison.png	results/baseline/figures/
policy_comparison_panel_K20_cap2.png	results/baseline/figures/
policy_comparison.png	results/archive/figures/optimization/

3. PROVENANCE ISSUES

● CRITICAL Issues

Issue 1: capacity_sensitivity_heatmap.png is BROKEN and NOT REPRODUCIBLE

- **Referenced in:** docs/core/technical_report.md:670 (Figure 6)
- **Location:** results/archive/figures/ (archive tier)
- **Problem:** The uploaded screenshot confirms the figure displays “Data format issue” for K=20 and K=40 panels — it shows blank heatmap panels with error text.
- **Root cause:** Both generate_capacity_sensitivity_heatmap.py and regenerate_all_figures.py read from results/capacity_comparison/ which **does not exist**. Data was relocated to results/analysis/capacity_comparison/ during directory restructuring.
- **The fix documented in** fleet_sensitivity_dual_investigation.md (POLICY_LABELS mapping) was applied to the script but the figure was never regenerated after the directory move.
- **Impact:** Figure 6 in the technical report is broken/misleading.

Issue 2: regenerate_all_figures.py has STALE PATHS — cannot reproduce 3 of 7 figures

- load_cbd_experiment() reads results/simulation/cbd_experiment/ → **does not exist** (actual: results/analysis/simulation/cbd_experiment/)
- load_capacity_comparison() reads results/capacity_comparison/ → **does not exist** (actual: results/analysis/capacity_comparison/)
- FIG_DIR = results/figures/ → but figures were redistributed to baseline/ , analysis/ , archive/
- **Affected figures:** fleet_sensitivity_dual.png (Fig 3), cbd_robustness_enhanced.png (Fig 4), capacity_sensitivity_heatmap.png (Fig 6)

⚠ MODERATE Issues

Issue 3: Technical report uses ARCHIVE figure when BASELINE equivalent exists

- **Figure 3** (fleet_sensitivity_dual.png): References results/archive/figures/ but results/baseline/figures/fleet_sensitivity_v2_dual.png exists with updated content.

- **Rationale concern:** Archive tier is documented as “Legacy (superseded)” but Figure 3 is a core result.

Issue 4: `policy_comparison_panel_K20_cap2.png` has NO identified generating script

- **Referenced in:** `docs/core/technical_report.md:433` (Figure 1)
- No script in `scripts/` or notebook in `notebooks/` contains code that generates `policy_comparison_panel_K20_cap2.png`.
- `docs/core/figure_trace_guide.md` attributes it to “Production analysis pipeline” — this is vague and non-reproducible.
- The figure exists and is valid (642 KB), but provenance is untraceable.

Issue 5: Multiple scripts have output path mismatches after directory restructuring

Script	Writes To	Figures Actually At
<code>analyze_queue_metrics.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>ana-lyze_seasonal_patterns.py</code>	<code>results/figures/</code>	<code>results/analysis/figures/</code>
<code>regenerate_all_figures.py</code>	<code>results/figures/</code>	<code>results/baseline/figures/</code> , <code>results/analysis/figures/</code> , <code>results/archive/figures/</code>
<code>generate_capacity_sensitivity_heatmap.py</code>	<code>results/figures/</code>	<code>results/archive/figures/</code>

Running any of these scripts would write to `results/figures/` (which now only contains a README) rather than the documented locations.

Issue 6: `cbd_response_comparison.png` , `cbd_scenario_comparison.png` , `cbd_heatmap.png` — notebook-only provenance

- These 3 figures (used in `docs/analysis/cbd_robustness_analysis.md`) are generated only by `notebooks/09_cbd_analysis.ipynb`.
- No standalone script exists to regenerate them, reducing reproducibility.

Issue 7: `visualization_index.md` references pre-restructure paths

- The quick reference table in `docs/core/visualization_index.md` lists most figures under `results/figures/` which is now essentially empty (only contains README).
- This documentation is stale and misleading.

4. RECOMMENDATIONS

Priority 1 — Fix Broken Figure (Critical)

1. **Regenerate** `capacity_sensitivity_heatmap.png` : Update `generate_capacity_sensitivity_heatmap.py` (and `regenerate_all_figures.py`) to read from `results/`

analysis/capacity_comparison/ instead of results/capacity_comparison/. Regenerate the figure and place it at the correct tier location. The technical report Figure 6 currently shows broken “Data format issue” panels.

Priority 2 — Fix Path Mismatches (High)

1. **Update all script output paths** after the directory restructuring:
 - regenerate_all_figures.py : Update FIG_DIR, load_cbd_experiment() path, load_capacity_comparison() path
 - analyze_queue_metrics.py : Change FIGURES_DIR to results/analysis/figures/
 - analyze_seasonal_patterns.py : Change FIGURES_DIR to results/analysis/figures/
 - generate_capacity_sensitivity_heatmap.py : Update both RESULTS_DIR and FIGURES_DIR

Priority 3 — Resolve Archive vs Baseline (Medium)

1. **Figure 3 (fleet_sensitivity_dual.png)**: Decide whether to:
 - (a) Update technical_report.md to reference the baseline version (fleet_sensitivity_v2_dual.png), or
 - (b) Regenerate fleet_sensitivity_dual.png from current production data and move to baseline
 - Either way, the archive figure should not be the canonical reference for a core report figure.

Priority 4 — Establish Provenance (Medium)

1. **policy_comparison_panel_K20_cap2.png** : Create a generating script or document the exact manual/ad-hoc process that produced this figure. As Figure 1 in the technical report, it must be reproducible.

Priority 5 — Documentation Cleanup (Low)

1. **Update visualization_index.md** : Reflect the new directory structure (baseline/figures/ , analysis/figures/ , archive/figures/).
2. **Create standalone scripts** for the 3 CBD figures currently only generated by notebook 09_cbd_analysis.ipynb .

5. AUDIT OF SPECIFICALLY REQUESTED FIGURES

fleet_sensitivity Figures

- **fleet_sensitivity_dual.png** (archive): Used as Figure 3 in technical report. Generated by regenerate_all_figures.py (but script has stale paths). Baseline has a different version (fleet_sensitivity_v2_dual.png). **Recommendation: Migrate to baseline.**
- **fleet_sensitivity_v2_dual.png** (baseline): Generated by production pipeline. Not referenced in any documentation.
- **fleet_sensitivity_curve.png** (archive): Not referenced in current docs.

capacity_sensitivity_heatmap

- **capacity_sensitivity_heatmap.png** (archive):  **BROKEN** — shows “Data format issue” in two of three panels. Referenced as Figure 6 in technical report. Scripts cannot regenerate due to stale directory paths. **Highest priority fix needed.**

Core Report Section Figures (technical_report.md)

Figure	File	Tier	Reproducible?
Figure 1	policy_comparison_panel_K20_cap2.png	baseline	✗ No script found
Figure 2	re-sponse_time_distribution_by_policy.png	baseline	⚠ Script path mismatch
Figure 3	fleet_sensitivity_du al.png	archive	⚠ Archive tier; script path mismatch
Figure 4	cbd_robustness_enhanced.png	analysis	⚠ Script path mismatch
Figure 5	cbd_equity_tradeoff_summary.png	analysis	⚠ Script path mismatch
Figure 6	capacity_sensitivity_heatmap.png	archive	🔴 BROKEN + not reproducible
—	queue_comparison_by_policy.png	analysis	⚠ Script path mismatch
—	season-al_patterns.png	analysis	⚠ Script path mismatch
—	dis-tance_comparison_bar.png	analysis	✅ OK

Only 1 of 9 figures referenced in the technical report is fully reproducible from current scripts.

Audit performed: 2026-03-21. Read-only — no files were modified.